

IndicConformer 600M

Multilingual ASR

Technical Research Document

A Comprehensive Study of AI4Bharat's
Multilingual Speech Recognition System
for 22 Official Indian Languages

Prepared for Academic Research

Topic: Multilingual Speech-to-Text

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1. Executive Summary

This document provides a comprehensive technical overview of the IndicConformer 600M Multilingual ASR (Automatic Speech Recognition) model developed by AI4Bharat. The model represents India's first open-source ASR system covering all 22 officially recognized Indian languages, achieving state-of-the-art performance through a novel Conformer-based hybrid architecture.

Key Highlights

- Model Size: 600 Million parameters

- La

2. Model Architecture

2.1 Conformer Architecture Overview

The Conformer architecture, introduced by Gulati et al. (2020), combines the strengths of Transformers and Convolutional Neural Networks (CNNs) for optimal speech recognition performance.

2.2 Conformer Block Components

Each Conformer block consists of four key modules stacked together:

1. Feed-Forward Network (FFN) 1: Half-step residual FFN module
- 2.

The model employs a hybrid decoding strategy with two approaches:

- CTC (Connectionist Temporal Classification): Faster inference, suitable for real-time applications
- RM

3. Supported Languages

IndicConformer 600M supports all 22 officially recognized Indian languages:

Language List

Language	Code	Language	Code
Assamese	as	Bengali	bn
Bodo	brx	Dogri	doi
Gujarati	gu	Hindi	hi
Kannada	kn	Konkani	kok
Kashmiri	ks	Maithili	mai
Malayalam	ml	Manipuri	mni
Marathi	mr	Nepali	ne
Odia	or	Punjabi	pa
Sanskrit	sa	Santali	sat
Sindhi	sd	Tamil	ta
Telugu	te	Urdu	ur

4. Performance Benchmarks

4.1 Hindi ASR Performance (Vistaar Benchmark)

The following table shows Hindi ASR performance on the Vistaar benchmark:

Dataset	Domain	Greedy WER	+ LM	Improvement
Kathbath	Read speech	10.34%	9.00%	-1.34%
Kathbath Noisy	Noisy read	11.86%	10.19%	-1.67%
FLEURS	Broadcast	12.68%	11.18%	-1.50%
CommonVoice	Crowd-sourced	16.57%	12.54%	-4.03%
IndicTTS	TTS-derived	9.49%	8.55%	-0.94%
MUCS	Conversational	10.41%	9.05%	-1.36%
Gramvaani	Rural/dialectal	27.61%	24.09%	-3.52%
Average		14.14%	12.09%	-2.05%

4.2 Multilingual Performance Comparison

Comparison of ASR Word Error Rate (WER %) across languages:

Language	IndicConformer	Whisper	Better
Hindi	11.2%	13.8%	IndicConformer
Bengali	12.4%	N/A	IndicConformer
Marathi	13.1%	N/A	IndicConformer
Tamil	14.6%	N/A	IndicConformer
Telugu	14.8%	N/A	IndicConformer
Kannada	16.2%	N/A	---
Average	13.7%	---	---

4.3 Inference Speed Performance

Real-Time Factor (RTF) analysis across different hardware:

Hardware	RTF	Speedup
Apple M4 - CPU	0.27x	3.7x faster
Apple M1-M4 - MPS	0.03-0.05x	20-30x faster
NVIDIA GPU (CUDA)	0.05-0.10x	10-20x faster

5. Advantages

1. Open-Source and Free: MIT license enables unrestricted use, modification, and distribution.

2.

6. Disadvantages and Limitations

1. Limited to 22 Languages: Only covers official Indian languages; no tribal/dialect support.

2.

7. Competition Analysis

7.1 Competitive Landscape

Comparison of IndicConformer with competing ASR systems:

Model	Languages	Params	Avg WER	License
IndicConformer	22	600M	13.7%	MIT
IndicWhisper	12	769M	13.6%	MIT
Whisper	99	1.5B	24%+	MIT
Google USM	100+	2B	N/A	Proprietary
Azure STT	10	N/A	~20%	Commercial
Google STT	10	N/A	~24%	Commercial
SraVaani	65	N/A	28.4%	Open

7.2 Feature Comparison

- vs Whisper: Better accuracy on Indian languages (13.7% vs 24%+), faster inference

- VS

8. Technical Implementation

8.1 Installation and Setup

Install required dependencies:

```
pip install transformers torch soundfile streamlit jiwer numpy
```

8.2 Basic Usage Example

```
from transformers import AutoModel
import torch, soundfile as sf

# Load model
model = AutoModel.from_pretrained(
    "ai4bharat/indic-conformer-600m-multilingual",
    trust_remote_code=True,
    token="YOUR_HF_TOKEN"
)

# Load audio
audio_data, sr = sf.read("audio.wav")
wav = torch.from_numpy(audio_data).float().unsqueeze(0)

# Transcribe
transcription = model(wav, "hi", "ctc")
print(transcription)
```

9. Evaluation Methodology

9.1 Metrics Used

Word Error Rate (WER)

$$\text{WER} = (S + D + I) / N * 100\%$$

Where S = Substitutions, D = Deletions, I = Insertions, N = Total words

Character Error Rate (CER)

$$\text{CER} = (S_c + D_c + I_c) / N_c * 100\%$$

Where S_c , D_c , I_c = Character-level errors, N_c = Total characters

Real-Time Factor (RTF)

$$\text{RTF} = t_{\text{processing}} / t_{\text{audio}}$$

RTF < 1: Faster than real-time, RTF > 1: Slower than real-time

9.2 Test Conditions

- Clean: Studio-quality, no noise

- No

10. Evaluated Results

10.1 Overall Performance

Metric	Value	Context
Avg WER (6 langs)	13.7%	Multilingual eval
Hindi WER	11.2%	Best performing
Hindi WER (with LM)	12.09%	With KenLM
Average CER	8.5%	Character-level
RTF (CPU)	0.27x	Apple M4
RTF (MPS)	0.03-0.05x	Apple Silicon GPU

10.2 Comparison with State-of-the-Art

System	Avg WER	Open Source
IndicConformer 600M	13.7%	Yes
IndicWhisper	13.6%	Yes
Nvidia NeMo large	18.6%	Yes
Azure STT	~20%	No
Google STT	~24%	No
SraVaani-1.0	28.4%	Yes

11. Confusion Analysis

11.1 Common Error Patterns

1. Phonetically Similar Words: Confusion between similar-sounding words, common in morphologically rich languages.

2.

Error Type	Percentage	Impact
Substitutions	62%	Most common error
Deletions	23%	Moderate impact
Insertions	15%	Least common

12. Research References

Primary References

- 1. Gulati et al., "Conformer: Convolution-augmented Transformer for Speech Recognition," Proc. Interspeech 2020 (arXiv: 2005.08100)

- 2.
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- 7. Rekesh et al., "Fast Conformer with Linearly Scalable Attention for Efficient Speech Recognition," IEEE SLT 2023

- 8.
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